

Programme: B E – Computer Science and Engineering (AI&ML)
Computer Science and Engineering (Cyber Security)
Internal Assessment – II

TERM : 03/03/2025-21/06/2025	COURSE NAME : AUTOMATA THEORY AND COMPILER DESIGN
DATE : 17-06-2025 TIME: – 3.00PM-4.00PM	COURSE CODE : CI44/CY44
MAX MARKS: 30	PORTIONS : L21-L40



Mobile Phones, Smart Watches and Electronic Gadgets are banned

Instructions to Candidates: Answer any two full questions.

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Construct the LL(1) Predictive Parsing Table for the following grammar $S \rightarrow a A B \mid b$ $A \rightarrow b B c \mid \epsilon$ $B \rightarrow a \mid \epsilon$	L3	3	5
1.b	Given the following syntax-directed translation rules, with <i>s</i> as synthesized attribute and <i>i</i> as inherited attribute; Rule 1: $R \rightarrow AB \{ B.i = R.i - 1; A.i = B.i; R.i = A.i + 1; \}$ Rule 2: $P \rightarrow CD \{ P.s = C.i + D.i; D.i = C.i + 2; \}$ Rule 3: $Q \rightarrow EF \{ Q.s = E.s + F.s; Q.s = F.i; \}$ Analyze the rules and justify your answer. i. Are the rules Valid? ii. Are the rules S-attributed or L-attributed?	L4	4	4
1.c	Consider the context free grammar given below, $P \rightarrow S$ $S \rightarrow \text{if} (B) S_1$ $B \rightarrow B_1 \parallel B_2 \mid B_1 \& \& B_2 \mid E_1 \text{ rel } E_2$ $E \rightarrow \text{id}$ $\text{rel} \rightarrow < \mid > \mid !=$ For the control statement: $\text{if} (a < b \& \& a != b) a = 0;$ i. Construct the annotated parse tree (Control-Flow Translation) with the help of translation scheme. ii. Generate the Three-Address Code (3AC)	L2, L3	5	6
2.a	Consider the Grammar $S \rightarrow \text{id} = E$ $E \rightarrow E + T \mid T$ $T \rightarrow T * F \mid F$ $F \rightarrow -E \mid \text{id}$ Construct DAG, Syntax Tree and Quadruple representation for the statements given i. $x = a + b * -c + b * -c$ ii. $x = a + b + b + a + a * b$	L3	5	6

2.b	Generate an L-attributed definition to compute the decimal value of a binary number considering the given CFG. Construct the Annotated Parse tree for the input Binary value 1110 = 14 Decimal equivalent. CFG: $L \rightarrow D L'$ $L' \rightarrow D L' \mid \epsilon$ $D \rightarrow 0 \mid 1$	L3	4	6
2.c	Consider the following grammar $S \rightarrow L = R \mid R$ $L \rightarrow * R \mid id$ $R \rightarrow L$ The LR(1) item sets are given below Canonical LR(1) parsing Identify LR(1) item sets that can be merged to construct LALR(1) item sets. Mention which sets were merged and why.	L4	3	3
3.a	Answer the following: - Given the expression statement $b = a[i] + c$. Convert the expression into three-address code (3AC). - Calculate the relative address and types for the declaration statement given with the help of annotated parse tree. $P \rightarrow D$ $D \rightarrow T id, D \mid \epsilon$ $T \rightarrow int \mid record \{ D \}$ int a; record { int b; int c; } d;	L4	5	6
3.b	Given the following grammar. Construct the LR (0) parse table CFG: $S \rightarrow n V n \mid p V p$ $V \rightarrow c$	L3	3	4
3.c	Construct Annotated Parse Tree and Dependency Graph for the following statements G: $D \rightarrow T L \mid T id C$ $T \rightarrow int \mid float$ $C \rightarrow [num] C_1 \mid \epsilon$ $L \rightarrow L_1, id \mid id$ - float a,b,c - int a[2][3]	L2, L3	4	5